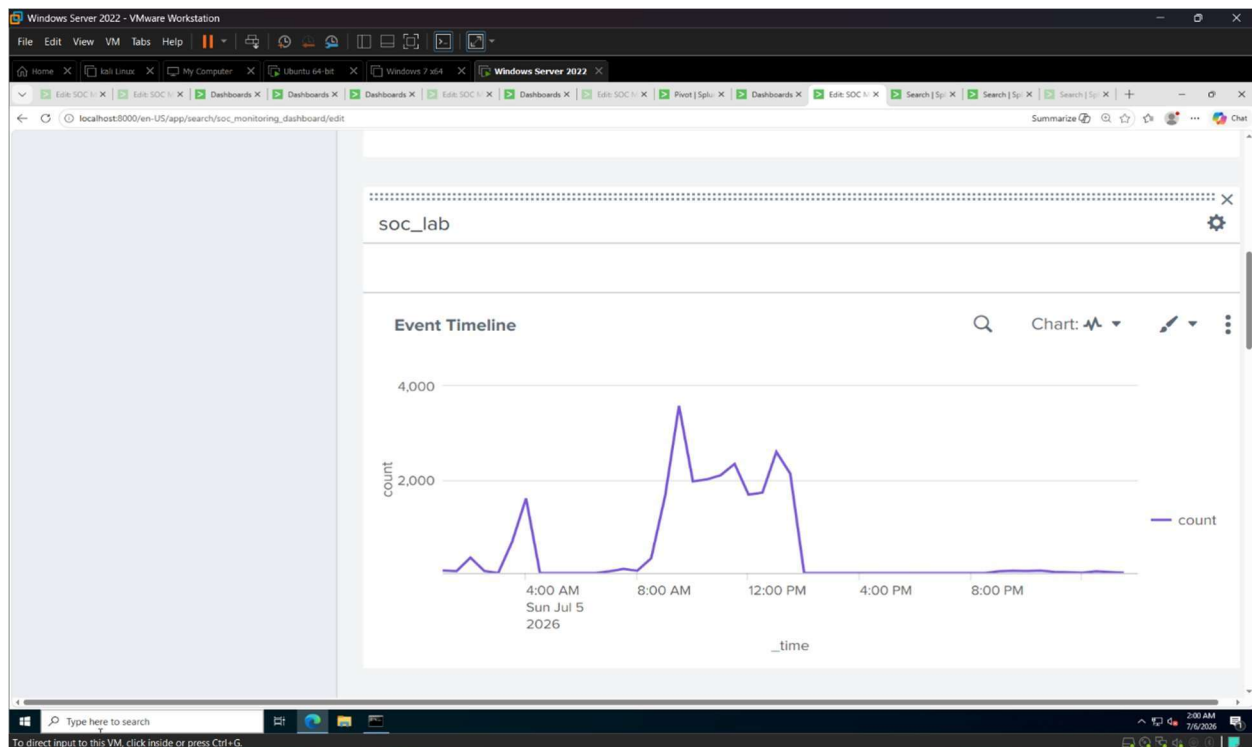




Insert Screenshot showing the Line Chart.

13.6 Task 4 – Add Windows Event Statistics

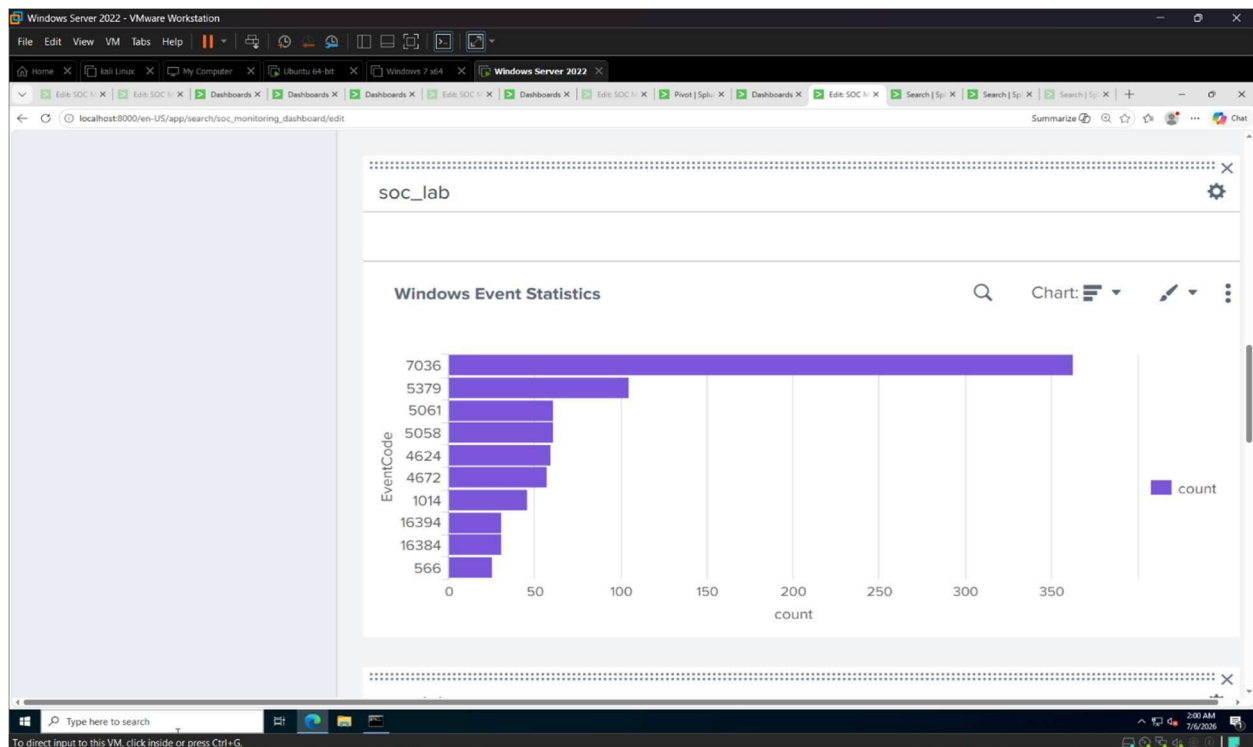
The following query displays the most common Windows Event Codes.

SPL Query `index=soc_lab sourcetype="WinEventLog:*"`

| top EventCode Visualization:

Bar Chart

Figure 13.4 – Windows Event Statistics



Insert Screenshot showing the Bar Chart.

13.7 Task 5 – Add Linux Syslog Panel

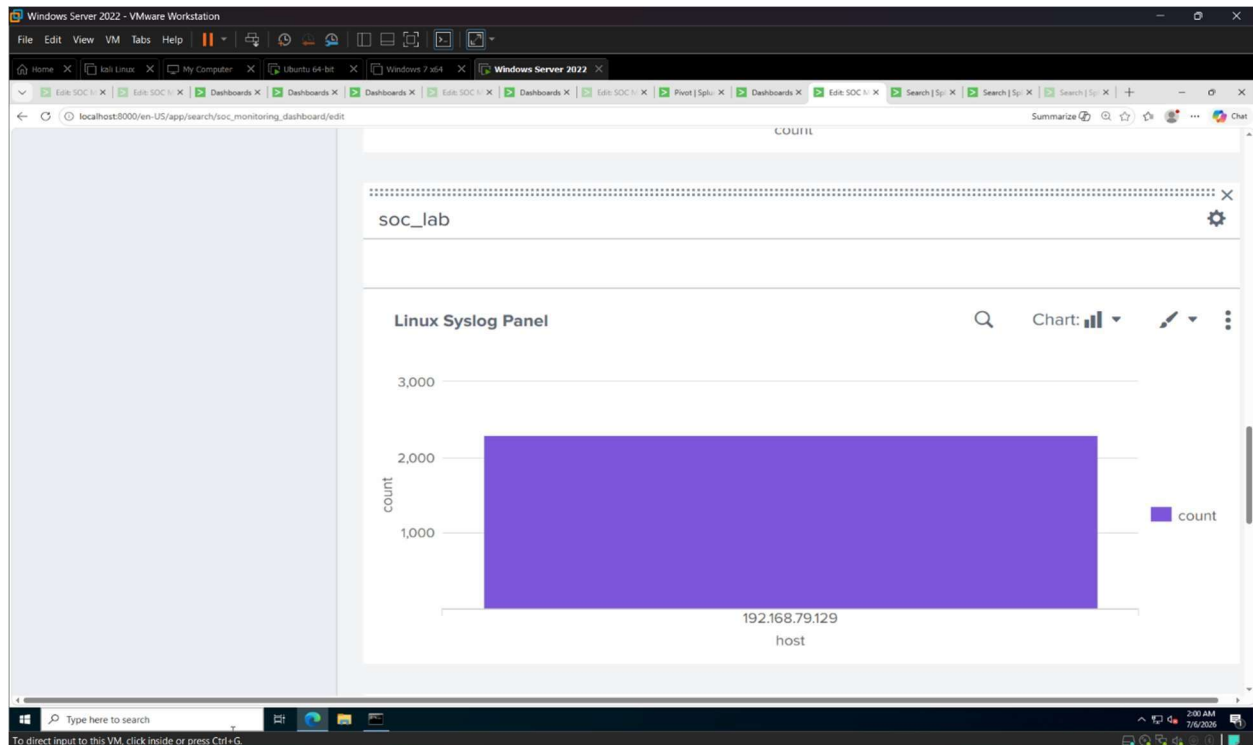
The following query displays Linux Syslog events.

SPL Query `index=soc_lab sourcetype=syslog`

| stats count by host Visualization:

Column Chart

Figure 13.5 – Linux Syslog Panel



Insert Screenshot showing Linux Syslog visualization.

13.9 Task 7 – Display Latest Security Events

The latest collected security events were displayed in table format.

SPL Query index=soc_lab

```
| table _time host sourcetype source
```

```
| sort - _time
```

Visualization:

Table

Figure 13.7 – Latest Security Events

The screenshot shows a Splunk dashboard with a table of security events. The table has the following data:

_time	host	sourcetype	source
2026-07-06 01:56:06.570	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:54:25.994	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:52:48.897	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:52:48.432	WIN-CIH5CI81EMR	WinEventLog:Application	WinEventLog:Application
2026-07-06 01:52:16.861	WIN-CIH5CI81EMR	WinEventLog:Application	WinEventLog:Application
2026-07-06 01:52:16.575	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:44:56.728	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:26:28.244	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:22:46.301	WIN-CIH5CI81EMR	WinEventLog:System	WinEventLog:System
2026-07-06 01:22:46.258	WIN-CIH5CI81EMR	WinEventLog:Application	WinEventLog:Application

Insert Screenshot showing the Table.

13.10 Tasks Performed

Created a custom Splunk dashboard.

Added a total event count panel.

Added an event timeline visualization.

Displayed Windows Event statistics.

Visualized Linux Syslog events.

Created a table showing the latest security events.

13.11 Summary

A custom Splunk dashboard was successfully created to visualize security events collected from Windows Server, Ubuntu Linux, and Windows Defender Firewall. Multiple dashboard panels were added using SPL queries to display event counts, timelines, Windows Event Codes, Linux Syslog activity, firewall events, and recent security logs. The dashboard provides a centralized monitoring interface that helps SOC analysts quickly identify security events and monitor the overall health of the lab environment.

14 – Attack Simulation

14.1 Introduction

Attack simulation is an important phase in a Security Operations Center (SOC) lab. It helps generate security events that can be collected, monitored, and analyzed using a SIEM platform such as Splunk Enterprise. By performing controlled attack simulations in a lab environment, SOC analysts can verify log collection, evaluate detection capabilities, and improve incident response procedures.

In this project, Kali Linux was used as the attacker machine, while Windows Server 2022 and Ubuntu Linux acted as target systems. Different attack simulations were performed to generate Windows Event Logs, Linux Syslog events, and network activities. The generated events were then verified in Splunk using the soc_lab index.

14.2 Objectives

The objectives of this chapter are:

Verify connectivity between Kali Linux and target machines.

Generate security events using controlled attack simulations.

Produce Windows and Linux logs.

Verify attack events in Splunk.

Prepare data for detection rule development.

14.3 Task 1 – Verify Network Connectivity

Before performing any simulation, connectivity between Kali Linux, Windows Server, and Ubuntu was verified.

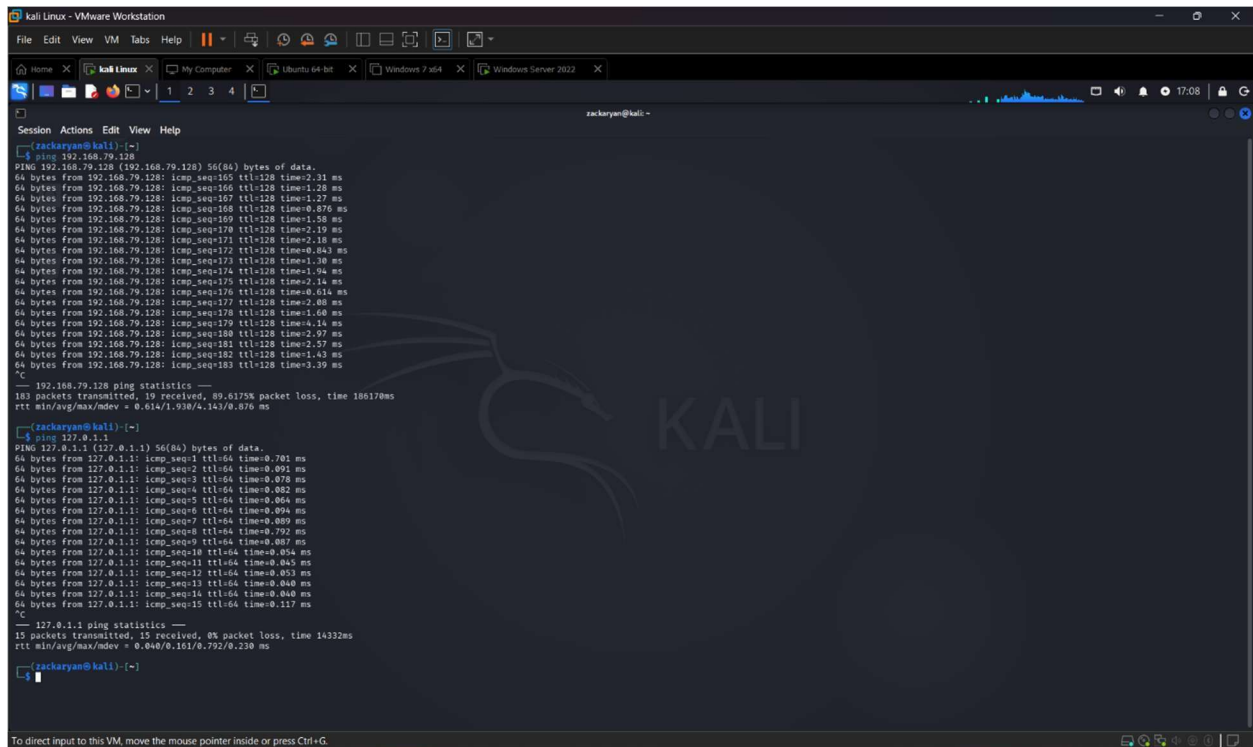
Commands

```
ping <Windows_Server_IP> ping
```

```
<Ubuntu_IP>
```

Successful replies confirmed that all virtual machines could communicate.

Figure 14.1 – Network Connectivity Test

The screenshot shows a Kali Linux terminal window with a dark background and a 'KALI' watermark. The terminal displays the output of two ping commands. The first command is 'ping 192.168.79.128', which shows 183 packets transmitted, 19 received, and an 89.6175% packet loss. The second command is 'ping 127.0.1.1', which shows 15 packets transmitted, 15 received, and 0% packet loss. The terminal window has a menu bar with 'File', 'Edit', 'View', 'VM', 'Tabs', and 'Help'. The title bar reads 'Kali Linux - VMware Workstation'. The terminal prompt is 'zackaryan@kali:~'.

Insert Screenshot showing successful ping results.

14.4 Task 2 – Perform Network Port Scan

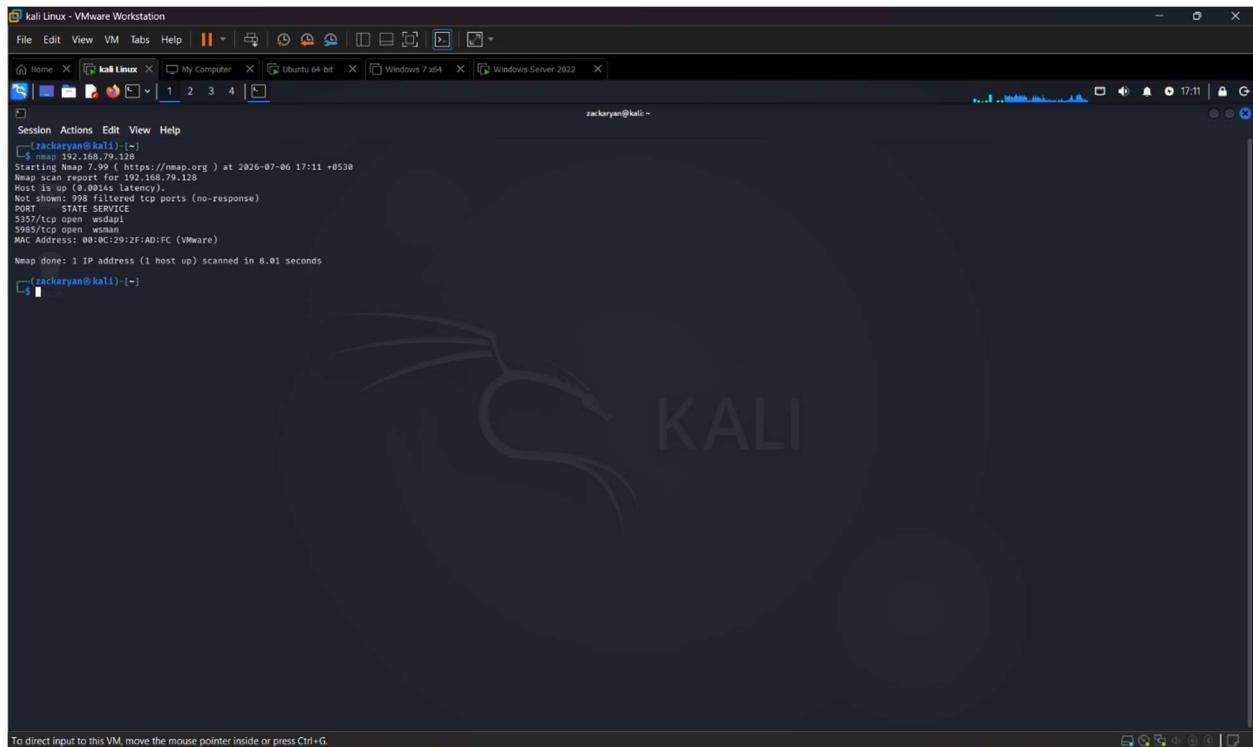
A network port scan was performed from Kali Linux to identify open ports on the Windows Server.

Command

nmap <Windows_Server_IP>

The scan displayed the open network ports and available services.

Figure 14.2 – Nmap Port Scan



Insert Screenshot showing the Nmap scan results.

14.5 Task 3 – Perform Service Version Detection

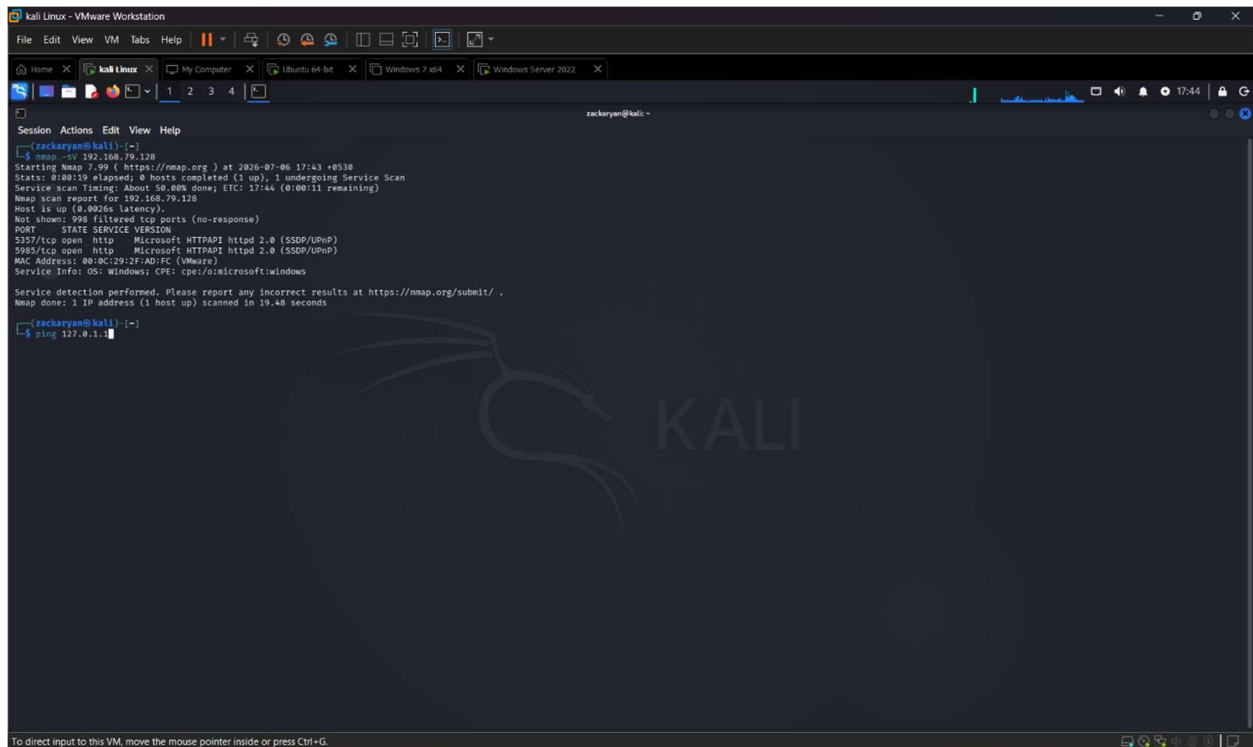
The services running on the Windows Server were identified.

Command

```
nmap -sV <Windows_Server_IP>
```

The command displayed service names and version information.

Figure 14.3 – Service Version Detection



Insert Screenshot showing the service version scan.

14.6 Task 4 – Generate Windows Login Events

Several login attempts were performed on the Windows Server using valid and invalid credentials to generate Windows Security Event Logs.

These login attempts created Security events such as successful logins and failed logins.

Figure 14.4 – Windows Login Simulation

Windows Server 2022 - VMware Workstation

File Edit View VM Jobs Help

Search | Splunk 10.4.1

localhost:8000/en-US/app/search/search?q=search%3Dindex%3Dsrc%3D%20sourcetype%3D%20WinEventLog%3A%20Security%3D%20EventCode%3D4625&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=

splunk> Enterprise

Search & Reporting

New Search

Index=soc_lab sourcetype=WinEventLog:Security EventCode=4625

Time range: Last 24 hours

5 events (7/5/26 5:00:00.000 AM to 7/6/26 5:27:37.000 AM) No Event Sampling

Events (5) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Time Event

7/6/26 5:05:34.725 AM LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-CIHC8181EHR host = WIN-CIHC8181EHR source = WinEventLog:Security sourcetype = WinEventLog:Security

7/6/26 5:05:30.388 AM LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-CIHC8181EHR host = WIN-CIHC8181EHR source = WinEventLog:Security sourcetype = WinEventLog:Security

7/6/26 5:05:25.875 AM LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-CIHC8181EHR host = WIN-CIHC8181EHR source = WinEventLog:Security sourcetype = WinEventLog:Security

7/6/26 5:05:21.995 AM LogName=Security EventCode=4625 EventType=0

File Explorer

Type here to search

To direct input to this VM, click inside or press Ctrl+G.

Windows Server 2022 - VMware Workstation

File Edit View VM Jobs Help

localhost:8000/en-US/app/search/search?q=search%3Dindex%3Dsrc%3D%20sourcetype%3D%20WinEventLog%3A%20Security%3D%20EventCode%3D4625&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=

splunk> Enterprise

Search & Reporting

New Search

Index=soc_lab sourcetype=WinEventLog:Security EventCode=4625

Time range: Last 24 hours

5 events (7/5/26 5:00:00.000 AM to 7/6/26 5:27:37.000 AM) No Event Sampling

Events (5) Patterns Statistics Visualization

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7/6/26 5:05:30.388 AM LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-CIHC8181EHR host = WIN-CIHC8181EHR source = WinEventLog:Security sourcetype = WinEventLog:Security

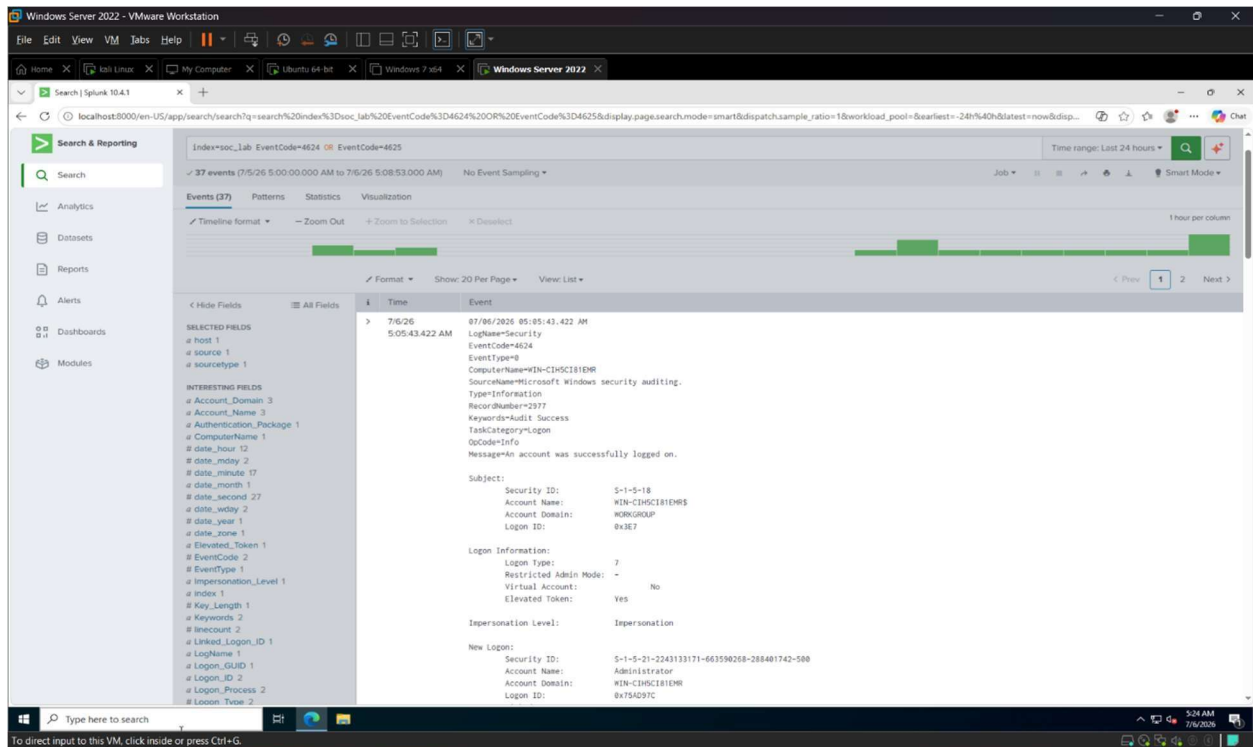
7/6/26 5:05:25.875 AM LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-CIHC8181EHR host = WIN-CIHC8181EHR source = WinEventLog:Security sourcetype = WinEventLog:Security

7/6/26 5:05:21.995 AM LogName=Security EventCode=4625 EventType=0

File Explorer

Type here to search

To direct input to this VM, click inside or press Ctrl+G.



Insert Screenshot showing Windows login activity or the resulting Security log events in Splunk.

14.7 Task 5 – Generate Linux Authentication Events

Authentication events were generated on the Ubuntu virtual machine.

Example activities included:

Logging into Ubuntu.

Executing administrative commands using sudo.

Running: logger "SOC Lab Attack

Simulation"

These activities produced Syslog events.

Figure 14.5 – Ubuntu Authentication Activity



Insert Screenshot showing the Linux terminal.

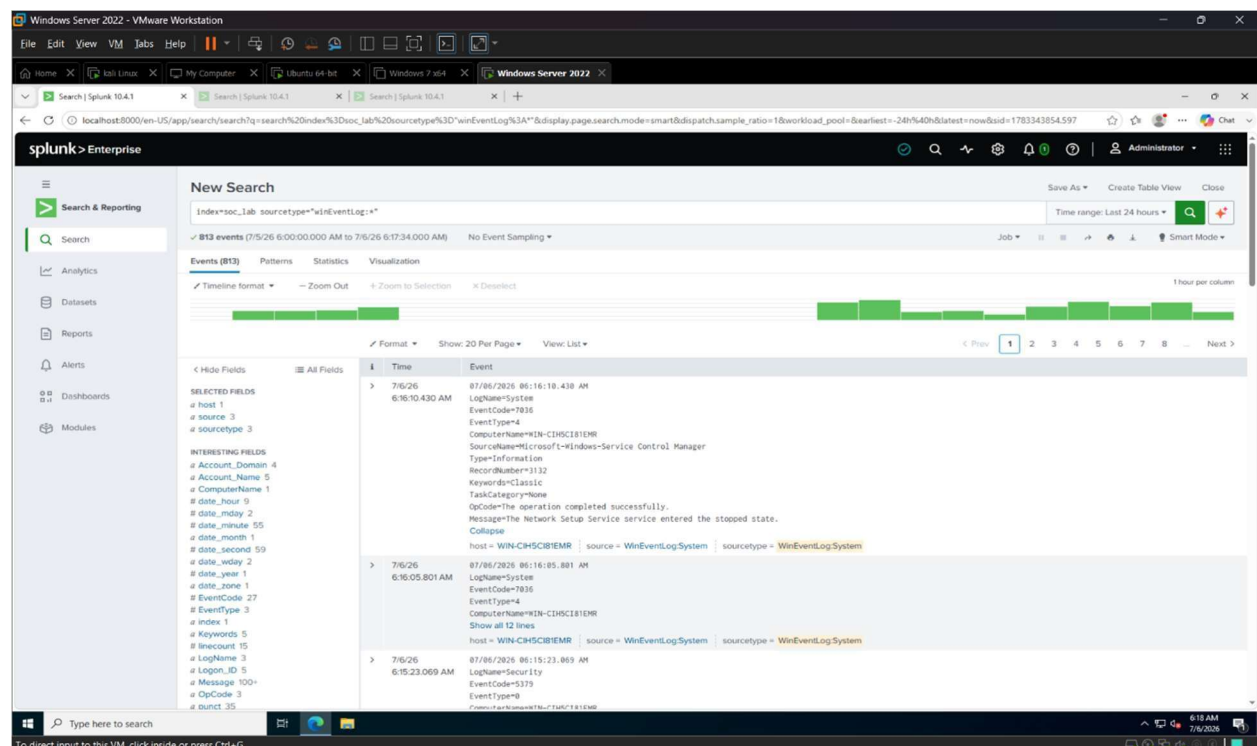
14.8 Task 6 – Verify Windows Events in Splunk The

generated Windows events were verified. SPL

Query

```
index=soc_lab sourcetype="WinEventLog:*
```

Figure 14.6 – Windows Events



Insert Screenshots showing Windows events.

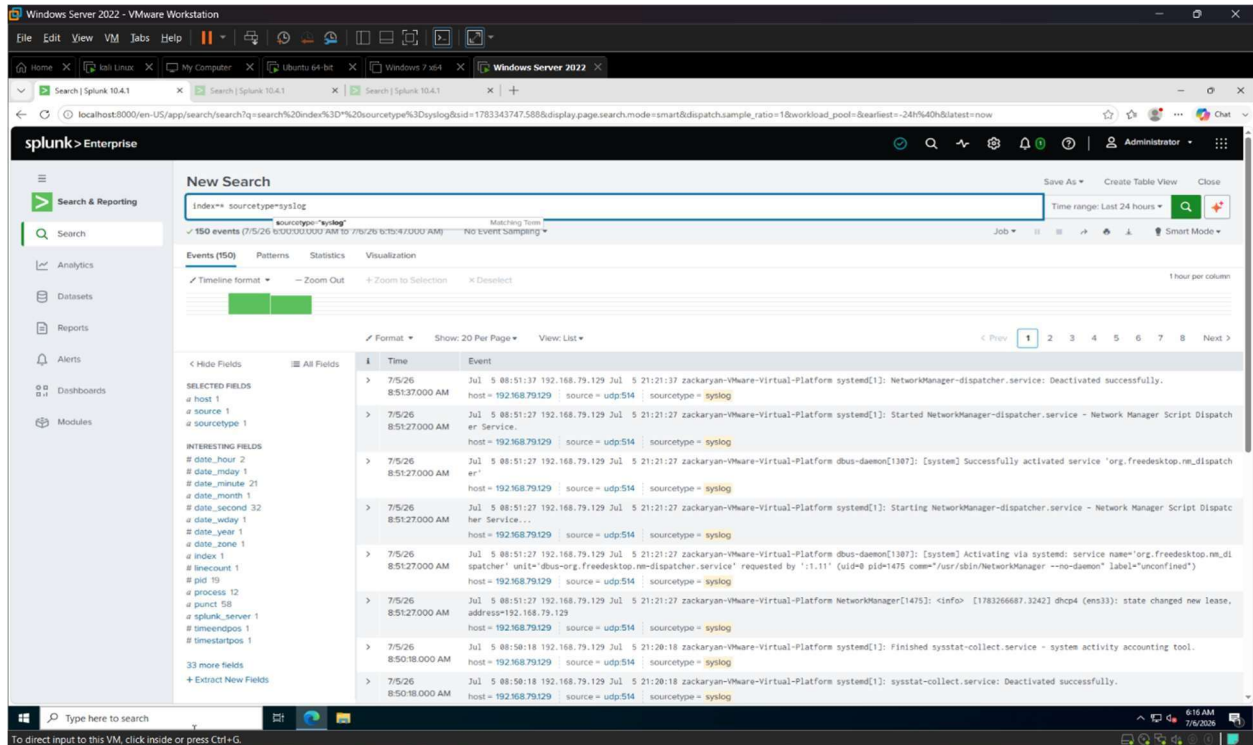
14.9 Task 7 – Verify Linux Events in Splunk

Linux Syslog events generated during the simulation were verified.

SPL Query

```
index=soc_lab sourcetype=syslog
```

Figure 14.7 – Linux Events

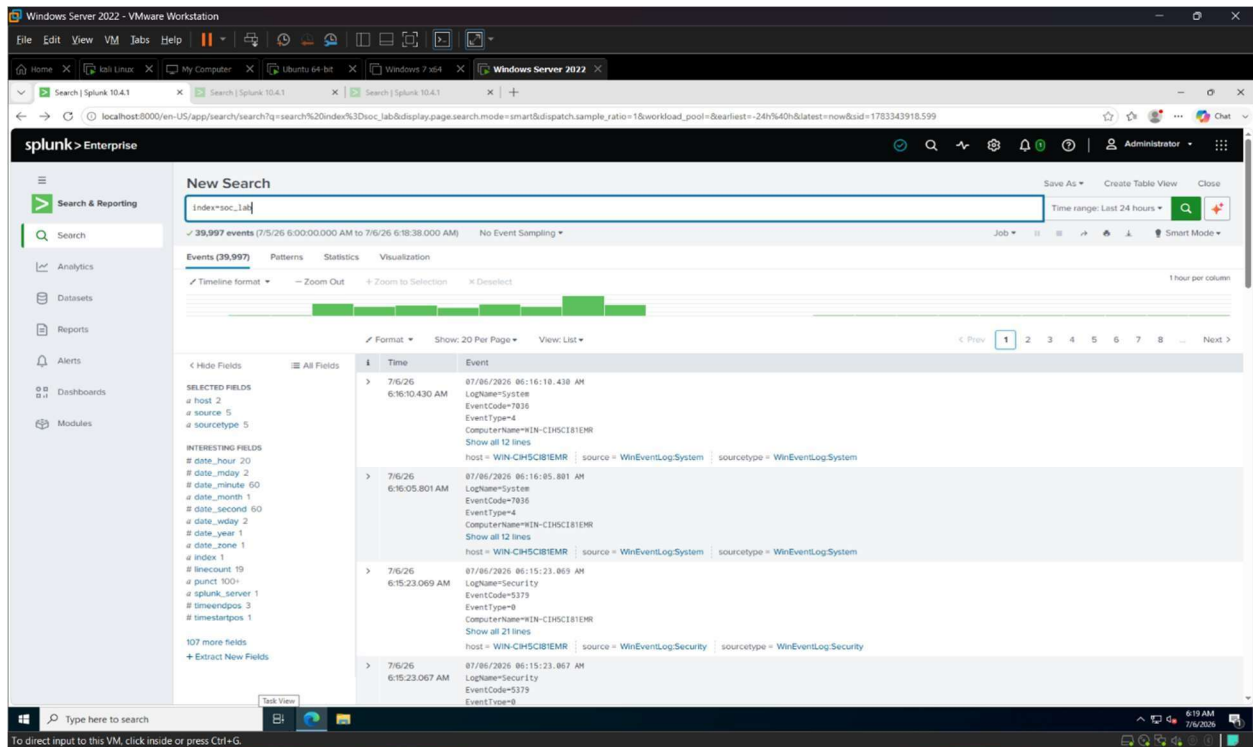


Insert Screenshot showing Linux Syslog events.

14.10 Task 8 – Verify All Generated Events

The following SPL query was executed to verify that all events generated during the attack simulation were successfully collected. SPL Query `index=s soc_lab`

Figure 14.8 – Attack Simulation Events



Insert Screenshot showing all generated events.

14.11 Tasks Performed

Verified communication between all virtual machines.

Performed a network port scan using Nmap.

Identified running network services.

Generated Windows Security events.

Generated Linux Syslog events.

Verified Windows logs in Splunk.

Verified Linux logs in Splunk.

Verified all generated events in the soc_lab index.

14.12 Summary

controlled attack simulations were successfully performed using Kali Linux within the SOC lab environment. Network connectivity was verified, port scanning activities were carried out, and Windows and Linux security events were generated. The resulting logs were successfully collected and verified in Splunk Enterprise. These events provide realistic

security data that will be used in the next chapter to develop and test security detection rules.

15 – Detection Rules

15.1 Introduction

Detection rules are one of the most important components of a Security Operations Center (SOC). They enable analysts to identify suspicious activities by continuously monitoring security events collected from different sources. In Splunk Enterprise, detection rules are created using Search Processing Language (SPL) queries that search for specific event patterns, authentication failures, and abnormal activities.

In this project, detection rules were developed using Windows Event Logs, Linux Syslog events, and Firewall logs stored in the soc_lab index. The rules were designed to detect common security events generated during the attack simulation phase.

15.2 Objectives

The objectives of this chapter are:

Create detection rules using SPL.

Detect suspicious Windows activities.

Detect Linux authentication events.

Detect firewall blocked connections.

Verify detection rule results.

15.3 Task 1 – Windows Failed Login Detection

A detection rule was created to identify multiple failed Windows login attempts.

SPL Query index=soc_lab EventCode=4625

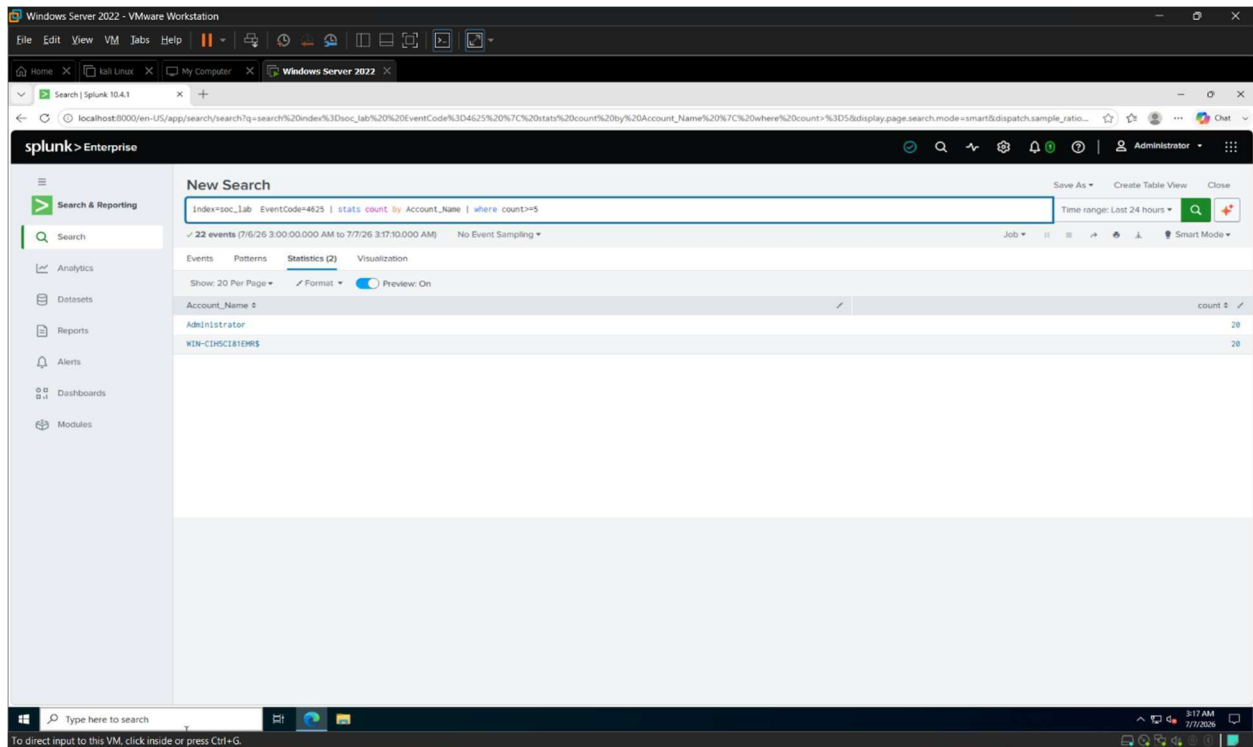
| stats count by Account_Name

| where count>=5

The rule identifies user accounts with five or more failed login attempts. Figure

15.1 – Windows Failed Login Detection

SUMIT KUMAR



Insert Screenshot showing the search results.

15.4 Task 2 – Windows Successful Login Detection

A rule was created to monitor successful Windows logins.

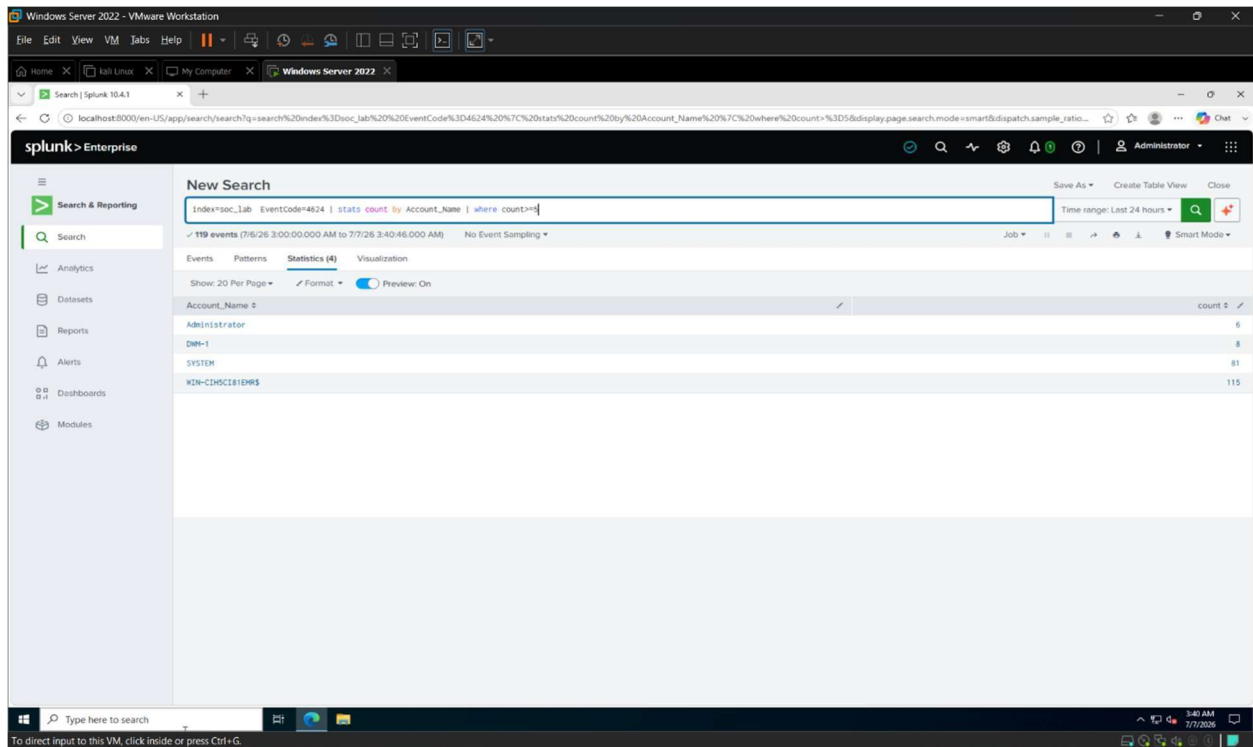
SPL Query `index=soc_lab EventCode=4624`

`| stats count by Account_Name`

`| where count>=5`

This query displays all successful login events.

Figure 15.2 – Successful Login Detection



Insert Screenshot showing successful login events.

15.5 Task 3 – Windows Account Lockout Detection

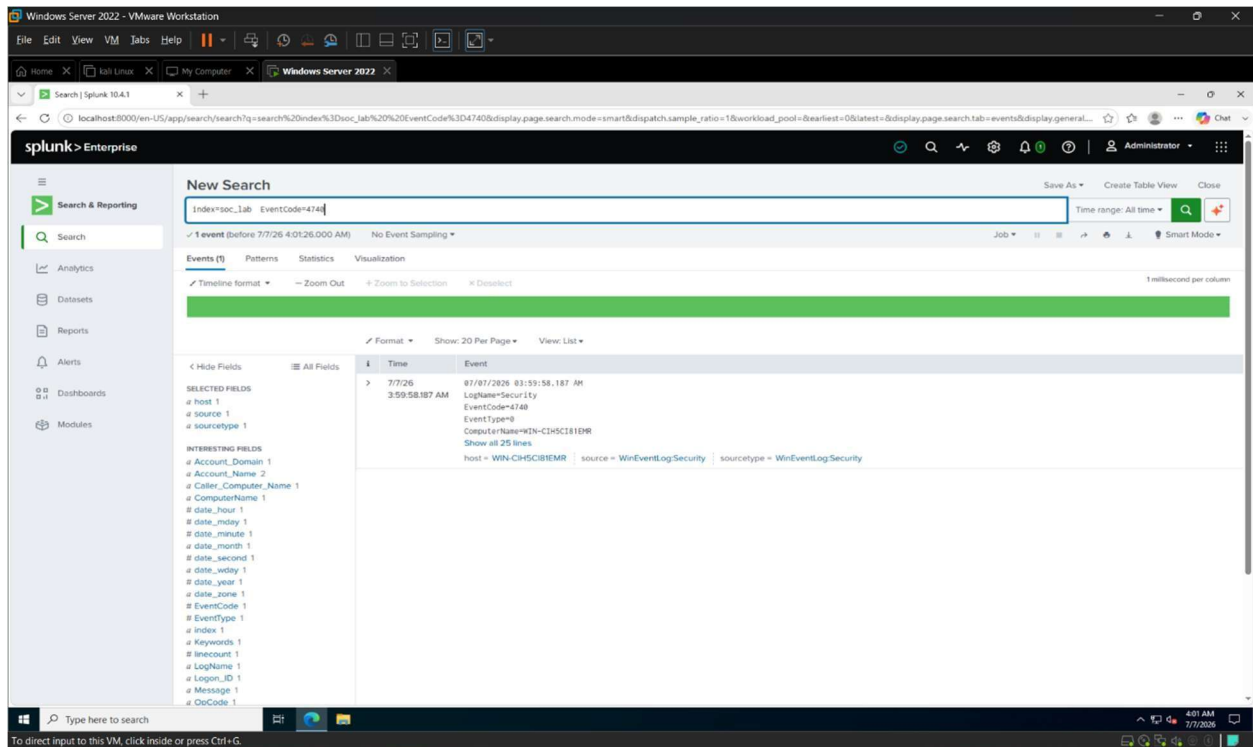
A rule was created to detect account lockout events. SPL

Query

`index=soc_lab EventCode=4740`

This query identifies accounts that have been locked due to multiple failed login attempts.

Figure 15.3 – Account Lockout Detection



Insert Screenshot showing Event ID 4740.

15.6 Task 4 – Linux Syslog Detection

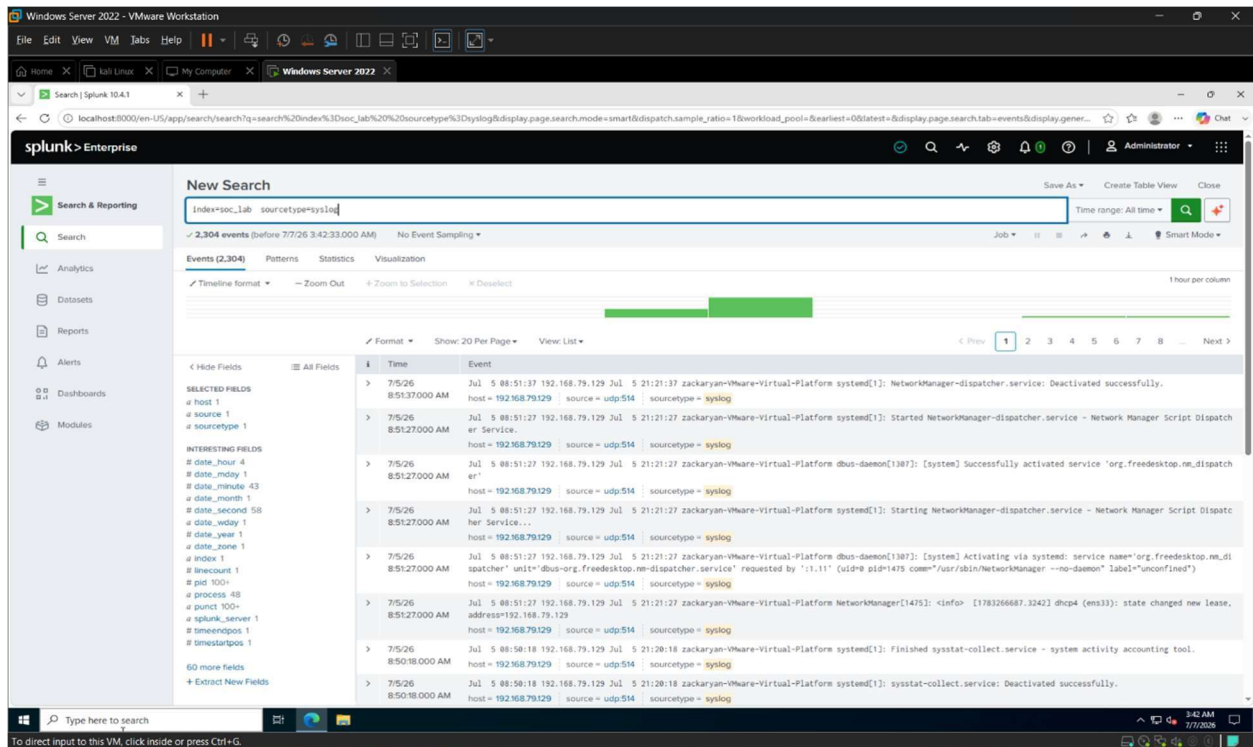
Linux authentication and system events were monitored using Syslog. SPL

Query

`index=soc_lab sourcetype=syslog`

This rule displays all Linux Syslog events received from the Ubuntu server.

Figure 15.4 – Linux Syslog Detection



Insert Screenshot showing Linux Syslog events.

15.7 Task 5 – Firewall Log Detection

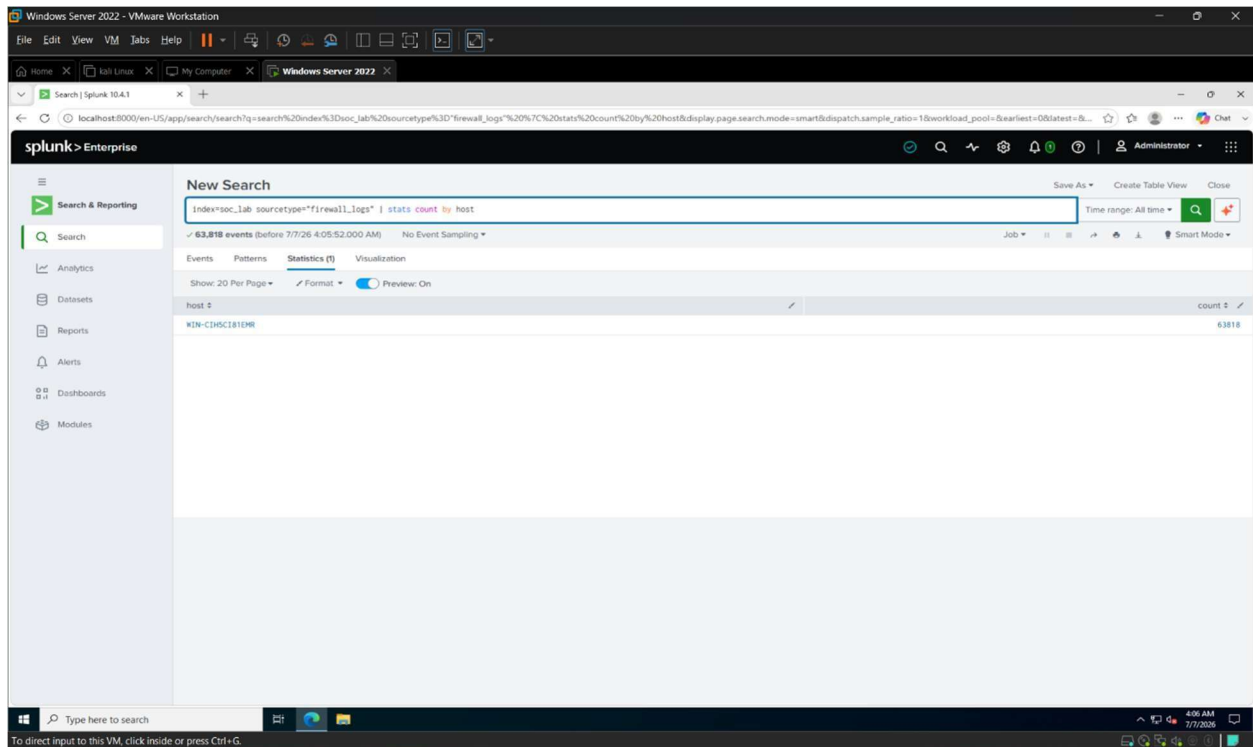
Firewall logs were analyzed to identify blocked network traffic.

SPL Query `index=soc_lab sourcetype=firewall_logs`

| stats count by host

The query displays the number of allowed and blocked firewall events.

Figure 15.5 – Firewall Detection



Insert Screenshot showing firewall events.

15.8 Task 6 – Event Trend Detection

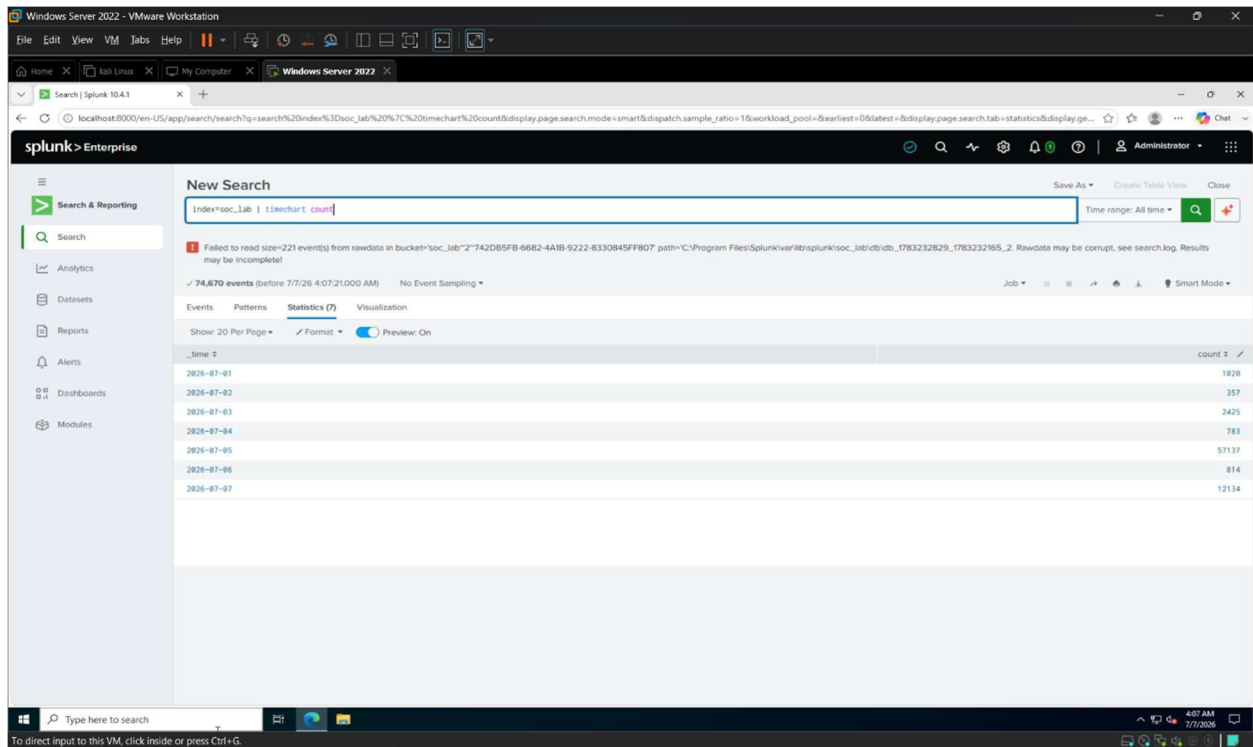
A time-based detection rule was created to monitor event activity.

SPL Query `index=soc_lab`

`| timechart count`

The visualization displays event trends over time.

Figure 15.6 – Event Trend Detection



Insert Screenshot showing the timechart.

15.9 Task 7 – Top Event Sources

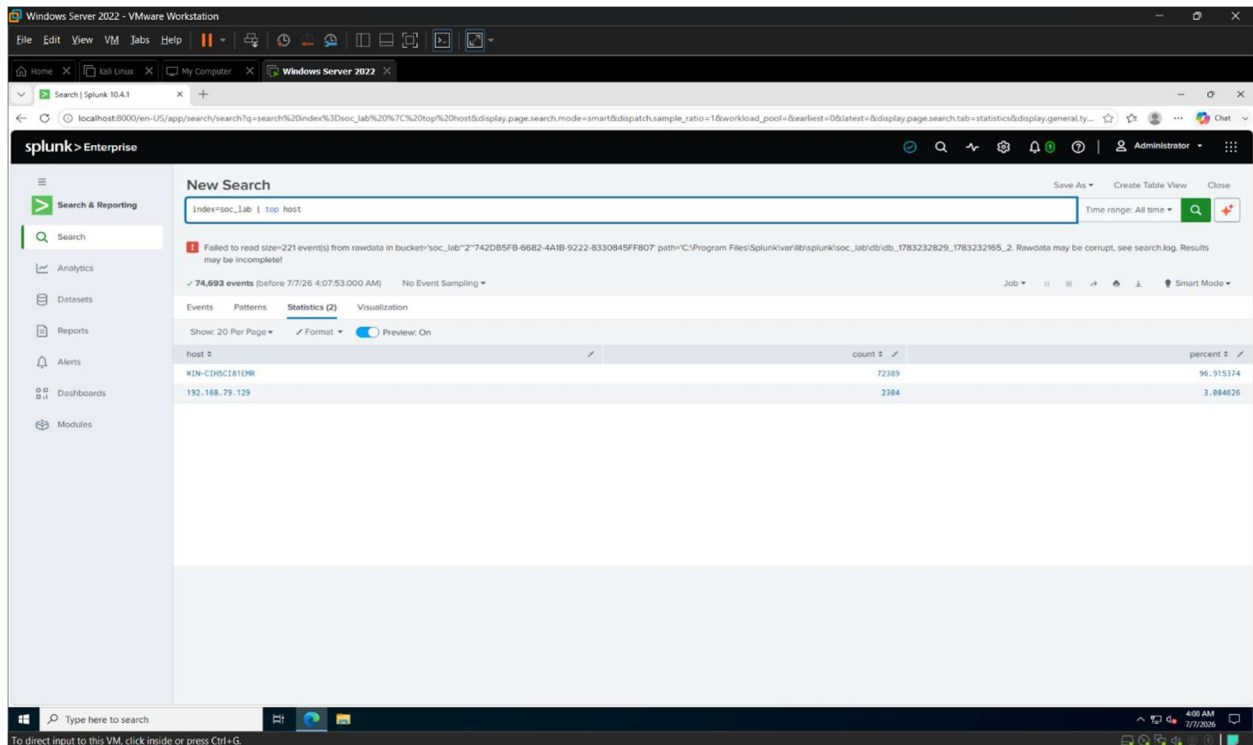
The following rule identifies the systems generating the highest number of logs.

SPL Query `index=soc_lab`

`| top host`

The output displays the hosts generating the most events.

Figure 15.7 – Top Event Sources



Insert Screenshot showing the top hosts.

15.10 Task 8 – Event Distribution by Sourcetype

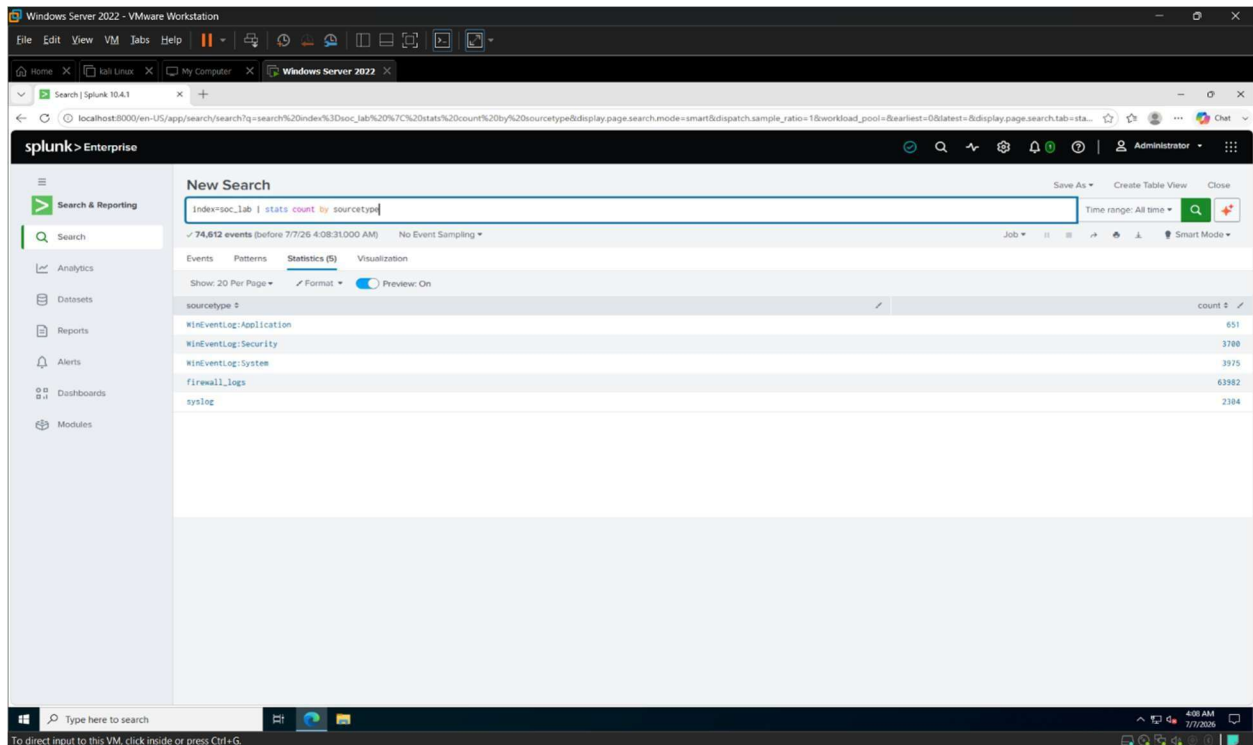
This rule categorizes collected events according to their source type.

SPL Query `index=soc_lab`

`| stats count by sourcetype`

The output shows the distribution of Windows, Linux, and Firewall logs.

Figure 15.8 – Event Distribution



Insert Screenshot showing event distribution.

15.11 Tasks Performed

Created Windows failed login detection rules.

Monitored successful Windows logins.

Detected Windows account lockout events.

Verified Linux Syslog events.

Monitored firewall logs.

Analyzed event trends over time.

Identified top event sources.

Classified logs based on source type.

15.12 Summary

Multiple detection rules were created using Splunk Search Processing Language (SPL). The rules successfully detected Windows authentication events, Linux Syslog activities, firewall events, and overall security trends. These detection rules demonstrate how Splunk

Enterprise can identify suspicious activities in a SOC environment and support security monitoring and incident detection.

16 – Results and Analysis

16.1 Introduction

The objective of this project was to build a functional Security Operations Center (SOC) monitoring environment using Splunk Enterprise. Windows Event Logs, Linux Syslog events, and Firewall logs were collected from different virtual machines and stored in the soc_lab index. Attack simulations were performed using Kali Linux to generate security events, which were then monitored, analyzed, and visualized in Splunk.

This presents the results obtained from the implemented SOC lab and analyzes the effectiveness of the configured monitoring solution.

16.2 Objectives

The objectives of this chapter are:

Analyze the collected security events.

Verify successful log collection.

Evaluate attack simulation results.

Review alert generation.

Analyze dashboard visualizations.

Summarize the overall SOC implementation.

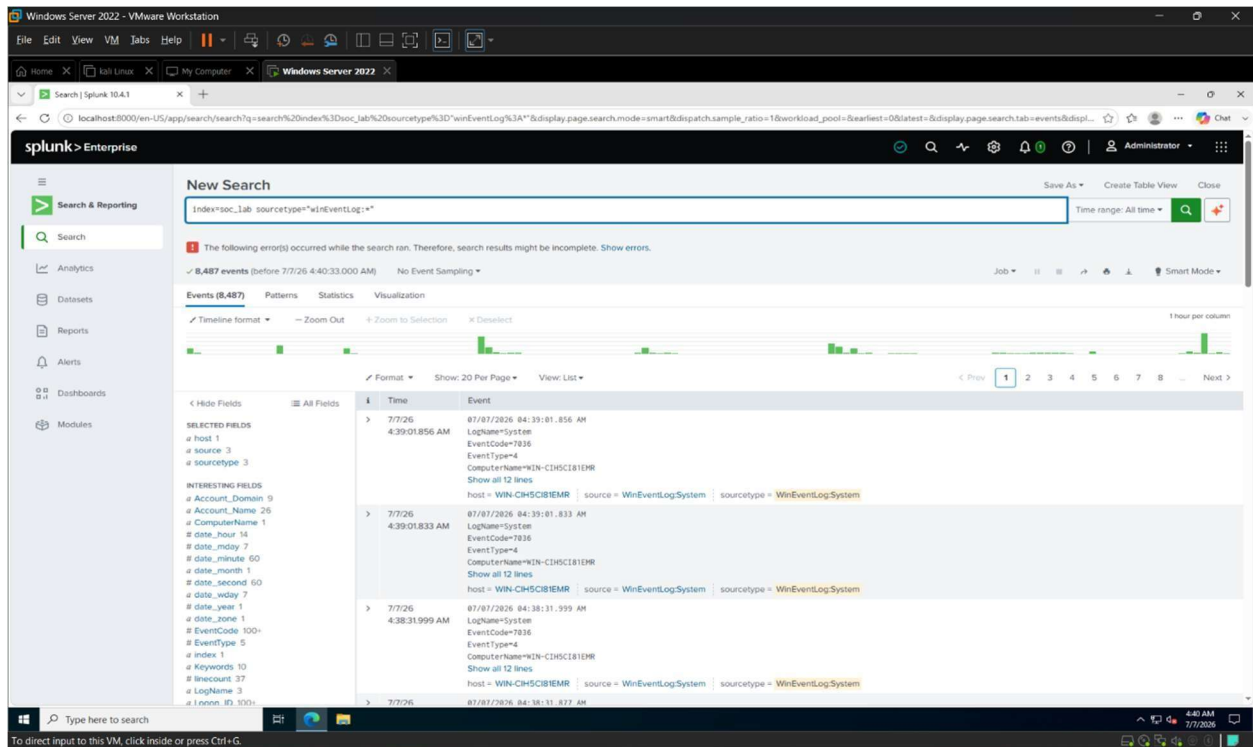
16.3 Result 1 – Windows Event Log Collection

Windows Event Logs were successfully collected from the Windows Server through the Splunk Universal Forwarder. Security, System, and Application events were indexed successfully in the soc_lab index. SPL Query

```
index=soc_lab sourcetype="WinEventLog:*
```

The search confirmed successful Windows log collection.

Figure 16.1 – Windows Event Logs



Insert Screenshot showing Windows Event Logs.

16.4 Result 2 – Linux Syslog Collection

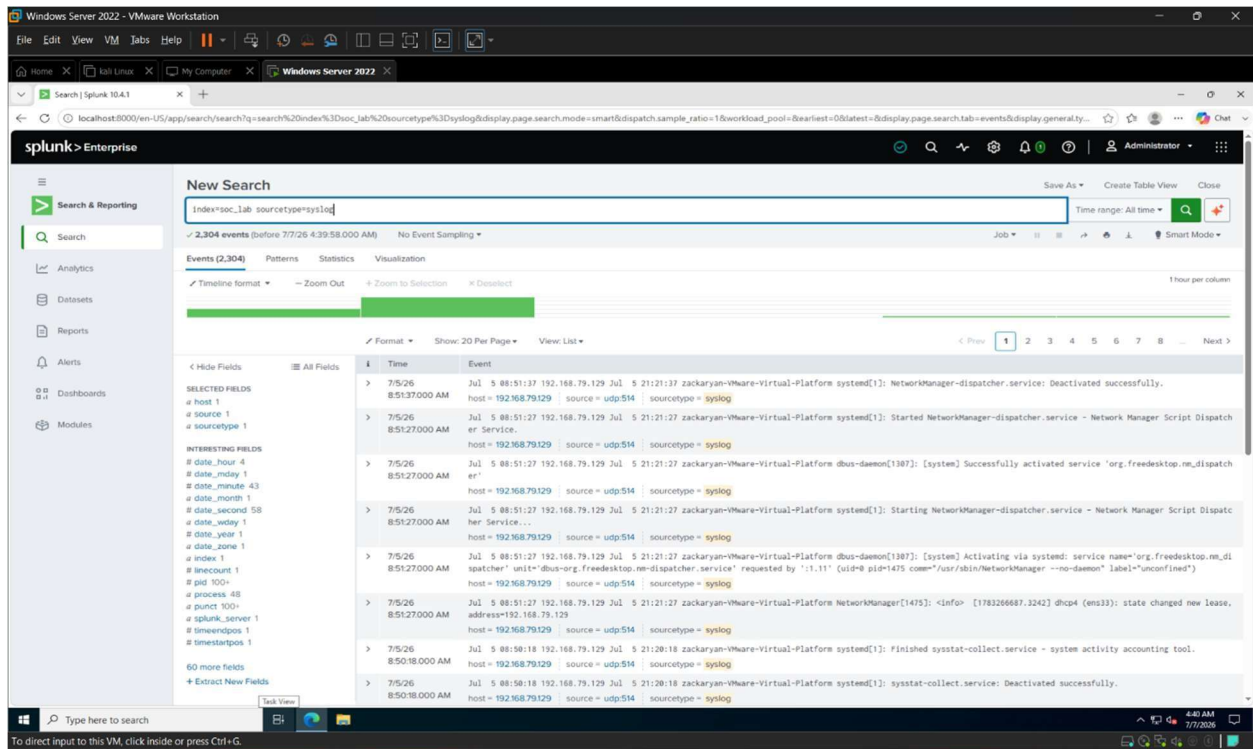
Ubuntu Linux successfully forwarded Syslog events to Splunk using the rsyslog service.

Authentication events, system messages, and manually generated logs were received successfully. SPL Query

`index=soc_lab sourcetype=syslog`

The received logs confirmed successful Syslog forwarding.

Figure 16.2 – Linux Syslog Events



Insert Screenshot showing Linux Syslog events.

16.5 Result 3 – Firewall Log Collection

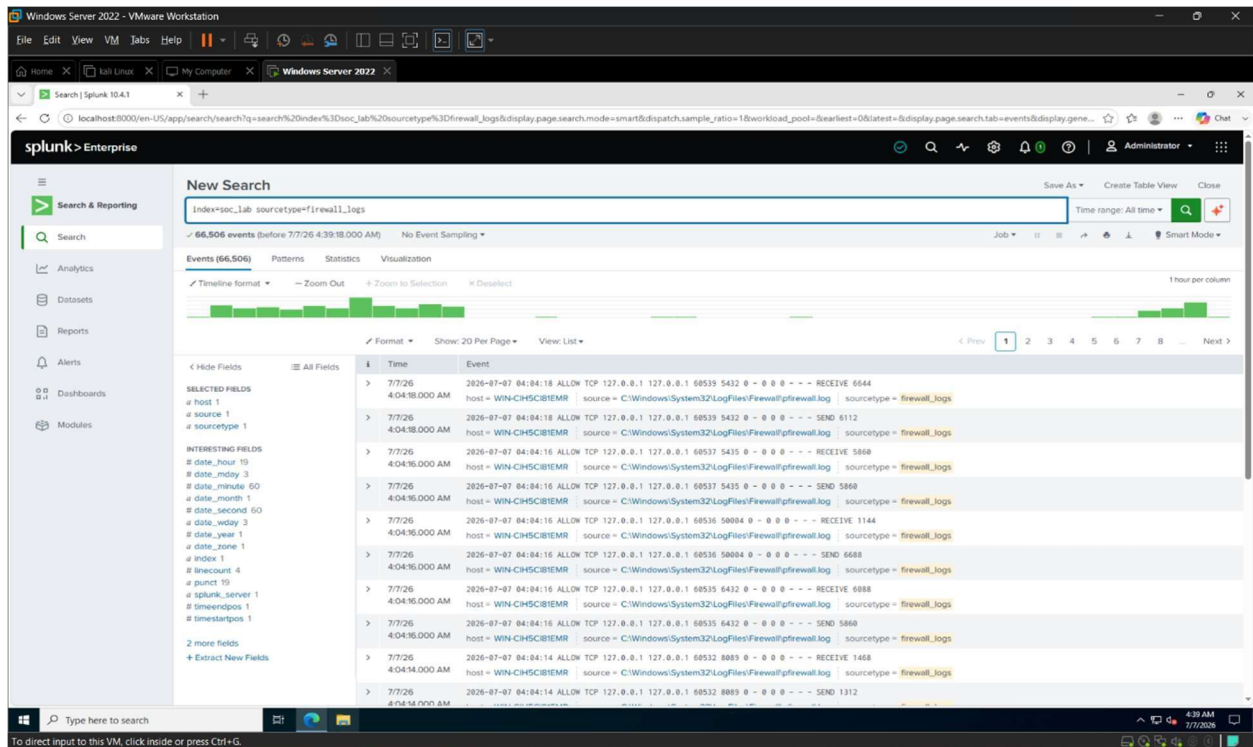
Firewall events were successfully collected and stored in Splunk Enterprise. SPL

Query

`index=soc_lab sourcetype=firewall_logs`

The collected firewall events verified successful log forwarding.

Figure 16.3 – Firewall Logs



Insert Screenshot showing Firewall logs.

16.6 Result 4 – Attack Simulation Analysis

Controlled attack simulations were performed from Kali Linux to generate security events.

The simulations included:

Network connectivity testing

Nmap port scanning

Windows login activity

Linux log generation

These activities successfully generated events that were indexed in Splunk Enterprise. 16.7

Result 5 – Alert Analysis

Security alerts configured in Splunk successfully detected matching events generated during testing.

The alerts demonstrated that the monitoring environment was capable of identifying

suspicious activities automatically. 16.8 Result 6 – Dashboard Analysis

Dashboard Studio successfully displayed security events using multiple visualizations.

The dashboard included:

Total Event Count

Windows Event Statistics

Linux Event Timeline

Firewall Activity

Latest Security Events

The dashboard provided a centralized and real-time monitoring interface.

16.9 Result 7 – Detection Rule Analysis

Detection rules successfully identified Windows authentication events, Linux Syslog activities, and Firewall events using SPL queries.

The rules demonstrated how Splunk can automatically detect security-related activities in a SOC environment.

16.10 Overall Analysis

The implementation of Splunk Enterprise successfully centralized log collection from multiple sources within the virtual SOC environment. Windows Server, Ubuntu Linux, and Firewall logs were indexed in the soc_lab index and analyzed using SPL queries. Attack simulations generated realistic security events that validated the effectiveness of log collection, detection rules, alerts, and dashboard visualizations.

The project demonstrated that Splunk Enterprise can efficiently support log management, event correlation, and security monitoring within a small-scale SOC lab.

16.11 Key Findings

Windows Event Logs were successfully collected.

Ubuntu Syslog forwarding operated correctly.

Firewall logs were indexed successfully.

Attack simulations generated useful security events.

Detection rules identified suspicious activities.

Alerts triggered correctly when configured conditions were met.

Dashboard Studio provided effective visualization of security data.

16.12 Summary

The results obtained from this project confirmed that Splunk Enterprise successfully collected, indexed, and analyzed security logs from multiple systems. Attack simulations validated the functionality of the monitoring environment, while detection rules, alerts, and dashboards provided effective visibility into security events. Overall, the implemented SOC lab achieved the project objectives and demonstrated the practical application of SIEM technology for centralized security monitoring.

17– Challenges and Troubleshooting

17.1 Introduction

During the implementation of the SOC lab, several technical challenges were encountered while configuring Splunk Enterprise, Windows Server, Ubuntu Linux, VMware Workstation, and Kali Linux. Most of these issues were related to virtual machine networking, log forwarding, firewall configuration, and data ingestion. Each issue was analyzed and resolved systematically to ensure the successful completion of the project.

17.2 Objectives

The objectives of this chapter are:

Identify the major challenges encountered during the project.

Explain the troubleshooting process.

Describe the solutions implemented.

Document lessons learned during implementation.

17.3 Challenge 1 – VMware Network Connectivity

Problem

The Windows Server and Ubuntu virtual machines were unable to communicate with each other. Network connectivity issues prevented log forwarding and remote communication.

Troubleshooting

Verified VMware network adapter settings.

Checked whether all virtual machines were connected to the same NAT network.

Verified IP address configuration using:

ipconfig ip

addr

Confirmed successful communication using the ping command.

Solution

After correcting the network configuration and assigning all virtual machines to the same NAT network, communication between Windows Server, Ubuntu, and Kali Linux was successfully established.

17.4 Challenge 2 – Universal Forwarder Not Sending Logs

Problem

Windows Event Logs were not appearing in Splunk Enterprise.

Troubleshooting

Verified Universal Forwarder installation.

Checked Forwarded Inputs.

Confirmed receiving port configuration (9997).

Restarted the Splunk Forwarder service.

Solution

After correcting the forwarding configuration, Windows Event Logs were successfully indexed in the soc_lab index.

17.5 Challenge 3 – Linux Syslog Not Received

Problem

Ubuntu Syslog events were not appearing in Splunk.

Troubleshooting

Verified rsyslog configuration.

Confirmed UDP port 514 was configured in Splunk.

Restarted the rsyslog service.

Tested connectivity using the logger command.

Commands used:

```
sudo systemctl restart rsyslog logger
```

"SOC Lab Test Message"

Solution

After correcting the Syslog forwarding configuration and enabling the UDP input, Linux logs were successfully received in Splunk.

17.6 Challenge 4 – Firewall Log Collection

Problem

Firewall logs were not initially available in Splunk.

Troubleshooting

Verified firewall logging configuration.

Checked Splunk Data Inputs.

Confirmed the correct source type and index.

Solution

After enabling firewall log collection and verifying the configuration, firewall events were successfully indexed.

17.7 Challenge 5 – Attack Simulation Verification

Problem

Attack simulation activities initially generated limited security events.

Troubleshooting

Verified connectivity between Kali Linux and target machines.

Repeated Nmap scans and authentication activities.

Confirmed that generated logs were indexed in Splunk.

Solution

After performing additional simulations and verifying the log sources, all required events were successfully collected and analyzed.

17.8 Lessons Learned

During the implementation of this project, the following key lessons were learned:

Proper network configuration is essential for communication between virtual machines.

Correct Splunk Data Input configuration is required for successful log collection.

Universal Forwarder must be configured correctly to forward Windows Event Logs.

Ubuntu Syslog forwarding depends on a properly configured rsyslog service.

Attack simulations are useful for validating detection rules, alerts, and dashboards.

Troubleshooting should always begin with network connectivity before checking application-level configurations.

17.9 Summary

Throughout the development of the SOC lab, several configuration and networking challenges were encountered. These issues were successfully resolved by verifying virtual machine networking, correcting Splunk configurations, enabling log forwarding, and validating attack simulations. The troubleshooting process improved the stability of the environment and ensured reliable log collection and security monitoring throughout the project.

18 – Best Practices

18.1 Introduction

Implementing best practices is essential for maintaining an effective and secure Security Operations Center (SOC) environment. Following standard procedures helps improve system performance, ensures reliable log collection, enhances security monitoring, and simplifies incident investigation.

During this project, several best practices were followed while configuring Splunk Enterprise, Windows Server, Ubuntu Linux, VMware Workstation, and Kali Linux. These practices helped maintain a stable and efficient SOC lab environment.

18.2 Objectives

The objectives of this chapter are:

Identify recommended practices for SOC implementation.

Improve log collection reliability.

Enhance security monitoring.

Maintain an organized Splunk environment.

Support efficient incident detection and analysis.

18.3 Best Practice 1 – Use a Dedicated Index

A dedicated index (soc_lab) should be used to store project-related logs instead of using the default index. This helps organize data, improves search performance, and simplifies log management.

18.4 Best Practice 2 – Configure Reliable Log Forwarding

All systems should be configured to forward logs correctly.

Windows Server should use Splunk Universal Forwarder.

Ubuntu Linux should use rsyslog.

Firewall logs should be forwarded using the appropriate data input.

Reliable log forwarding ensures that security events are continuously collected without interruption.

18.5 Best Practice 3 – Monitor Data Inputs Regularly

Splunk Data Inputs should be reviewed periodically to ensure they remain active and continue receiving logs from all configured sources.

Recommended checks include:

UDP Inputs

TCP Inputs

Forwarded Inputs

Receiving Port Configuration

Regular monitoring helps identify data collection issues at an early stage.

18.6 Best Practice 4 – Use Meaningful SPL Queries

Search Processing Language (SPL) queries should be optimized and focused on specific security events instead of performing unnecessary broad searches.

Examples include:

Failed login attempts

Successful logins

Firewall events

Linux Syslog events

Optimized queries improve search performance and make investigations more efficient.

18.7 Best Practice 5 – Configure Detection Rules and Alerts

Detection rules and alerts should be configured for important security events such as:

Multiple failed login attempts

Account lockouts

Authentication failures

Firewall blocked connections

Suspicious system activities

Timely alerts help SOC analysts respond quickly to potential security incidents.

18.8 Best Practice 6 – Use Dashboards for Continuous Monitoring

Dashboard Studio should be used to create centralized dashboards that display:

Total Events

Windows Event Logs

Linux Syslog Events

Firewall Activity

Recent Security Events

Dashboards provide real-time visibility into the monitored environment.

18.9 Best Practice 7 – Maintain Proper Network Configuration

All virtual machines should remain connected to the same VMware Workstation NAT network to ensure stable communication.

Regular verification should include:

IP Address

Network Adapter Status

Ping Connectivity

DNS Configuration

Proper networking is critical for successful log forwarding

18.10 Best Practice 8 – Perform Regular Testing

Periodic testing should be performed to verify that the monitoring environment continues to function correctly.

Recommended tests include:

Network connectivity tests

Log forwarding verification

Attack simulations

Alert testing

Dashboard verification

Regular testing ensures that configuration changes do not impact security monitoring.

18.11 Best Practice 9 – Keep Documentation Updated

All configuration changes, SPL queries, alert settings, and troubleshooting steps should be properly documented.

Well-maintained documentation simplifies maintenance, troubleshooting, and future enhancements.

18.12 Best Practices Followed During the Project

During the implementation of this SOC lab, the following practices were followed:

A dedicated soc_lab index was created.

Windows Event Logs were collected using Splunk Universal Forwarder.

Ubuntu Syslog events were forwarded using rsyslog.

Firewall logs were successfully integrated.

Detection rules and alerts were configured.

Dashboard Studio was used for visualization.

Attack simulations were performed using Kali Linux.

Log collection was verified after every major configuration change.

18.13 Summary

This chapter presented the best practices followed during the implementation of the SOC monitoring environment. Proper log collection, optimized SPL queries, reliable data forwarding, effective alert configuration, dashboard visualization, and regular testing significantly improved the efficiency and reliability of the SOC lab. Following these practices helps maintain a stable monitoring environment and supports effective security operations.

19– Conclusion

19.1 Introduction

This project was carried out to design and implement a functional Security Operations Center (SOC) monitoring environment using Splunk Enterprise. The project focused on collecting security logs from multiple systems, monitoring events, performing attack simulations, creating alerts, developing dashboards, and analyzing security data in a centralized platform.

The implementation demonstrated the practical use of a Security Information and Event Management (SIEM) solution for security monitoring within a virtual lab environment.

19.2 Project Achievements

The following objectives were successfully achieved during the project:

Splunk Enterprise was successfully installed and configured.

Windows Event Logs were collected using Splunk Universal Forwarder.

Ubuntu Linux Syslog events were forwarded using rsyslog.

Firewall logs were integrated into Splunk.

Security events were stored in the soc_lab index.

Multiple SPL queries were executed to analyze collected logs.

Controlled attack simulations were performed using Kali Linux.

Security alerts were configured and tested.

Detection rules were created for monitoring suspicious activities.

A centralized SOC Monitoring Dashboard was developed using Dashboard Studio. 19.3

Overall Outcome

The implemented SOC monitoring environment successfully centralized logs from Windows Server, Ubuntu Linux, and Firewall sources. Splunk Enterprise provided efficient log collection, indexing, searching, visualization, and alerting capabilities.

The project demonstrated how security events generated during attack simulations could be detected, analyzed, and monitored through dashboards and detection rules. The implementation also improved understanding of SIEM architecture, log management, and practical SOC operations.

19.4 Learning Outcomes

During this project, the following practical skills were developed:

Splunk Enterprise installation and configuration.

Windows Event Log collection.

Linux Syslog forwarding.

Firewall log integration.

Search Processing Language (SPL).

Alert configuration.

Dashboard development using Dashboard Studio.

Attack simulation using Kali Linux.

Security event monitoring and analysis.

Basic SOC operations and troubleshooting.

19.5 Summary

This project successfully demonstrated the implementation of a small-scale SOC monitoring environment using Splunk Enterprise. Security logs from multiple platforms were collected, analyzed, and visualized through dashboards, while alerts and detection rules improved security monitoring capabilities. The project achieved all defined objectives and provided practical experience in SIEM deployment, log analysis, and SOC operations.